Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Deep Learning, Production ML Systems, Anomaly Detection | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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19th October 2026

Re: Machine Learning Engineer – TD (Layer 6)

Dear Hiring Manager,

I am applying for the Machine Learning Engineer role at Layer 6, TD's AI center of excellence in Toronto. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, and I build end-to-end ML systems that run in production under hard constraints. Layer 6 ships models that reach millions of customers, and that is the standard I want to work to: clean code, reliable systems, and results judged in deployment.

Three parts of your posting match my work directly. For production deep learning, my published real-time pattern detection system runs at 14 ms on a Raspberry Pi 4, 74x faster than the DTW baseline, at F1 0.76 (Journal of the Audio Engineering Society, 2026), and I owned its full lifecycle from data acquisition to deployment and monitoring. For data-intensive financial work, at Forestpin I designed ML and statistical pipelines for financial forensics and compliance monitoring on large structured enterprise datasets, with SQL and Python scoring services for anomaly flagging in production. For scale and ownership, at MAS Holdings I led ML and computer vision systems for manufacturing, built real-time ingestion and ETL pipelines for high-frequency IoT data, and set up code review and CI/CD practices; the systems cut inspection time by 99.5% and raised operational efficiency by up to 300%. I work in Python and C++ with PyTorch, TensorFlow and JAX, and I have trained models on GPUs, including LoRA fine-tuning of a generative audio model across two GPUs.

I should be direct about the gaps. I have not used LangGraph or built Java systems, and my banking-scale transaction and document work is limited to the Forestpin period, so my Gen AI agent experience is lighter than my time-series and deep learning experience. I have built the same pieces with RAG, LLMs and HuggingFace, and I expect to pick up your stack quickly. I also have a research record, with 10+ peer-reviewed publications and four open datasets on Zenodo.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately for onsite work in Toronto. I would welcome the opportunity to discuss how my experience can support the machine learning team at TD (Layer 6). My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD